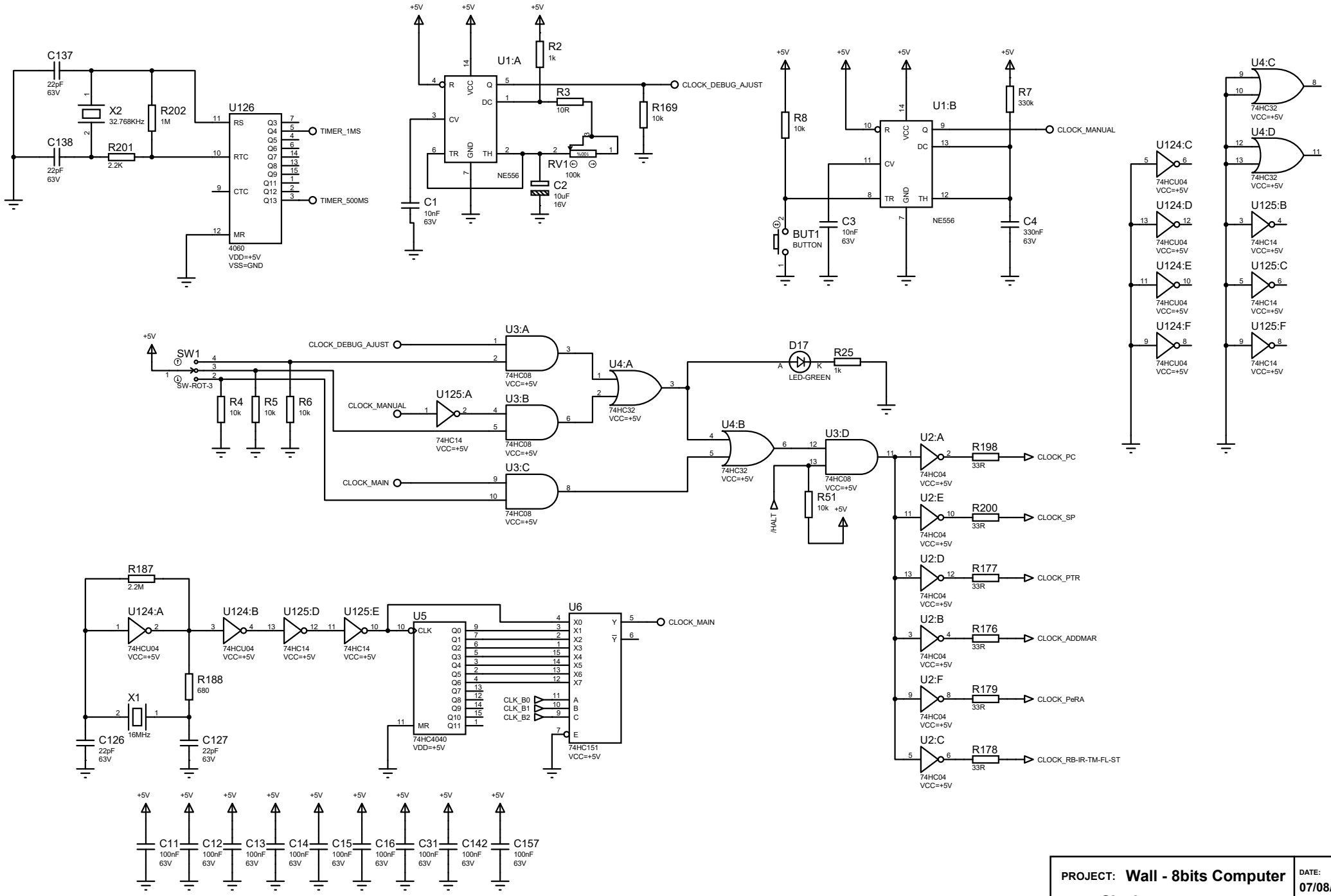
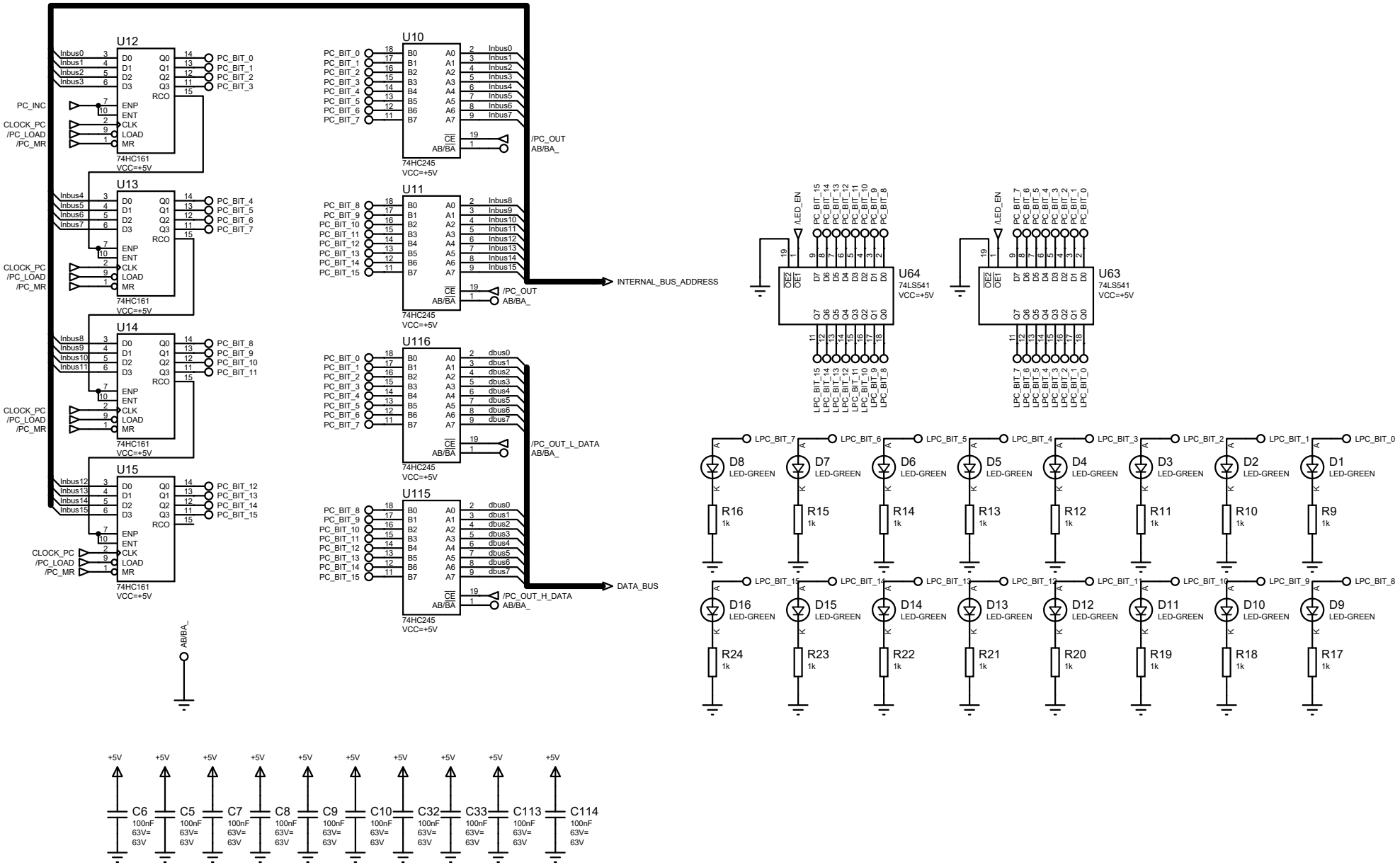
CLOCK



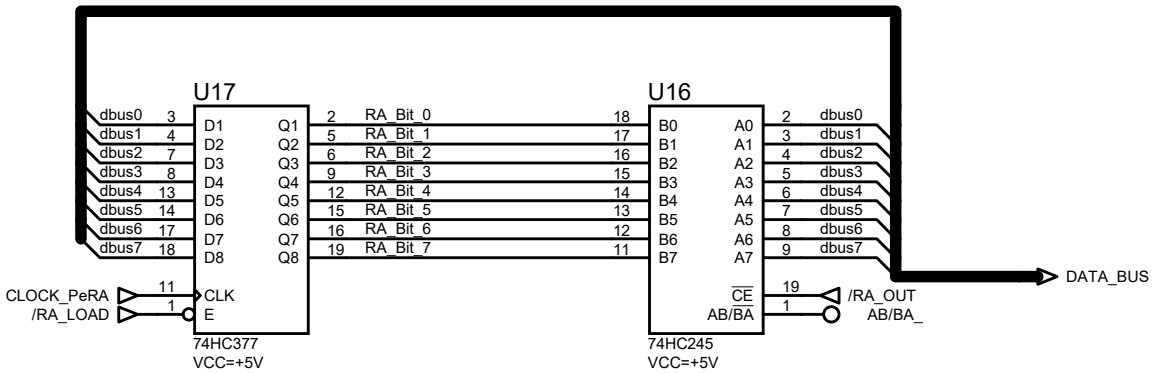
PROJECT: **Wall - 8bits Computer**		DATE: **07/08/2026**
PAGE: **Clock**		
FILE NAME: **Hardware.pdsprj**		PAGE: **1 of 17**
DRAWN BY: **Vinicius Jesus**	REV: **1.0**	

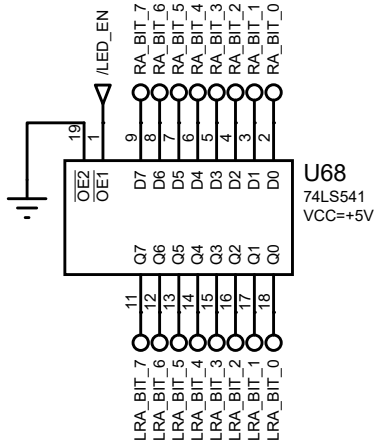
PC - Program Counter

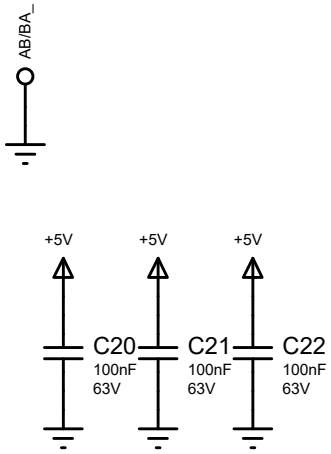


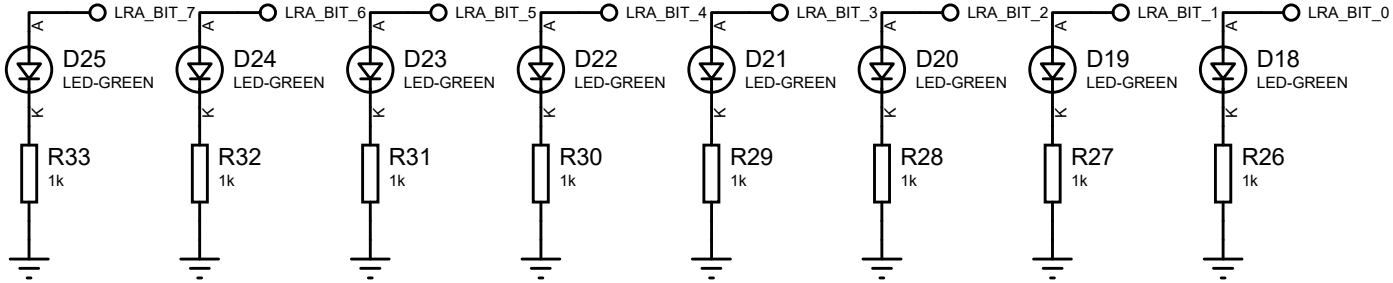
PROJECT: **Wall - 8bits Computer**	DATE: **07/08/2026**
PAGE: **Program Counter 16bits**	PAGE: **2 of 17**
FILE NAME: **Hardware.pdsprj**	
DRAWN BY: **Vinicius Jesus** REV: **1.0**	

Register A



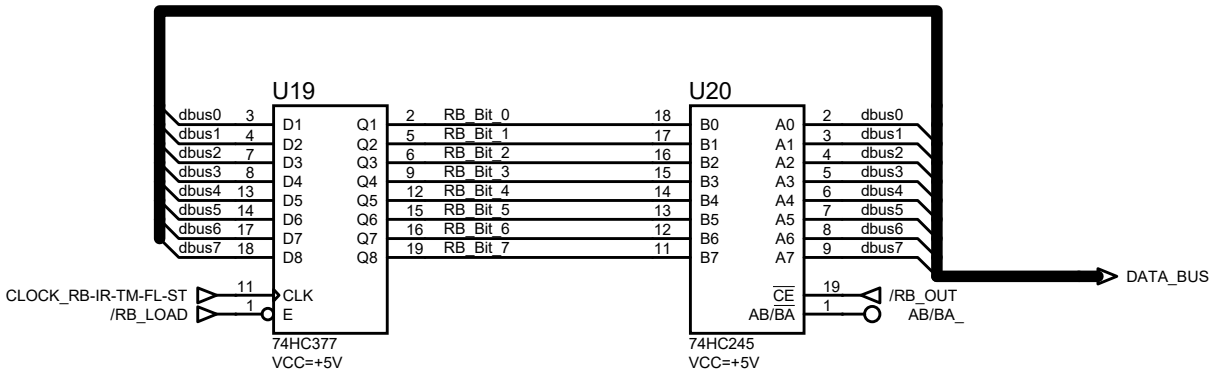


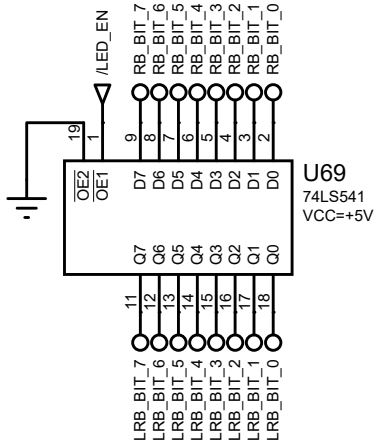


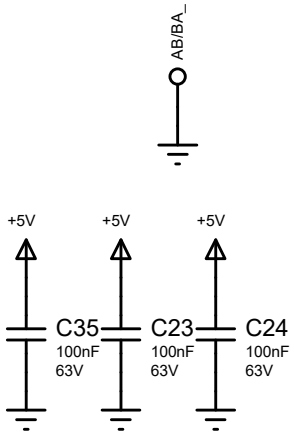


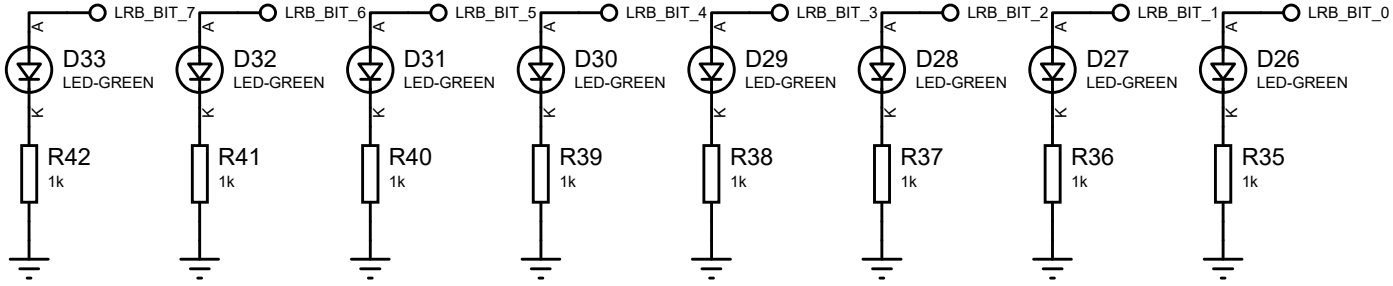
PROJECT: Wall - 8bits Computer	**DATE:** 07/08/2026
PAGE: Registrador A , ACC	
FILE NAME: Hardware.pdsprj	**PAGE:** 3 of 17
DRAWN BY: Vinicius Jesus **REV:** 1.0	

Register B



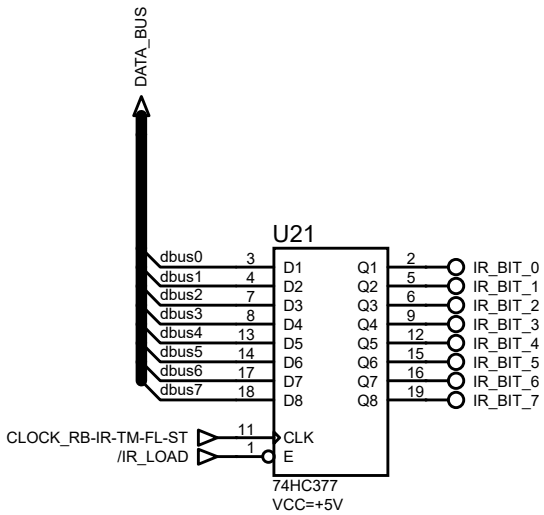


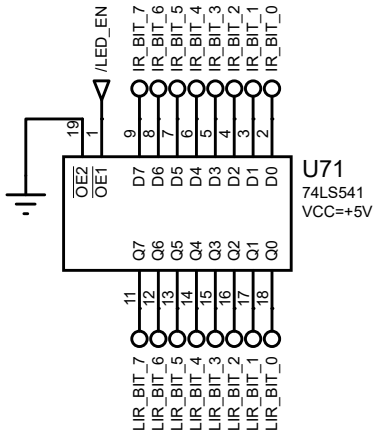


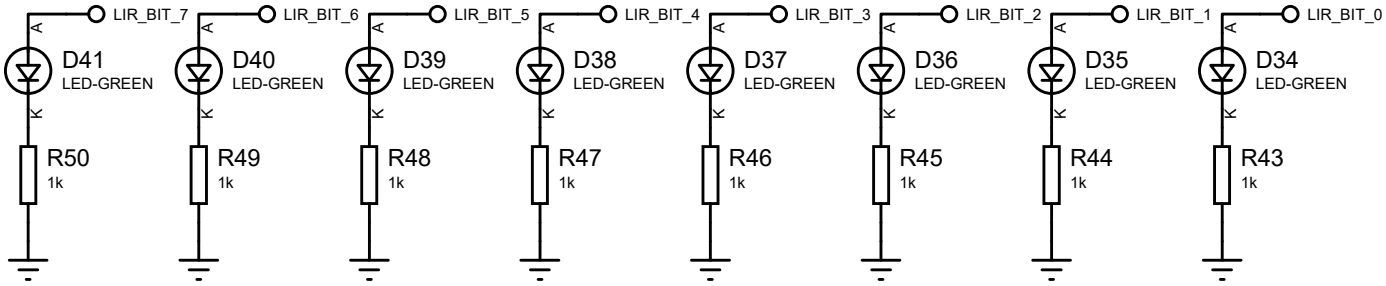


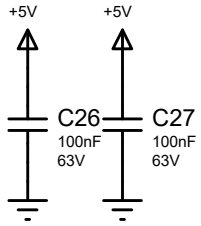
PROJECT: Wall - 8bits Computer	**DATE:** 07/08/2026
PAGE: Registrador B	
FILE NAME: Hardware.pdsprj	**PAGE:** 4 of 17
DRAWN BY: Vinicius Jesus **REV:** 1.0	

IR - Instruction Register





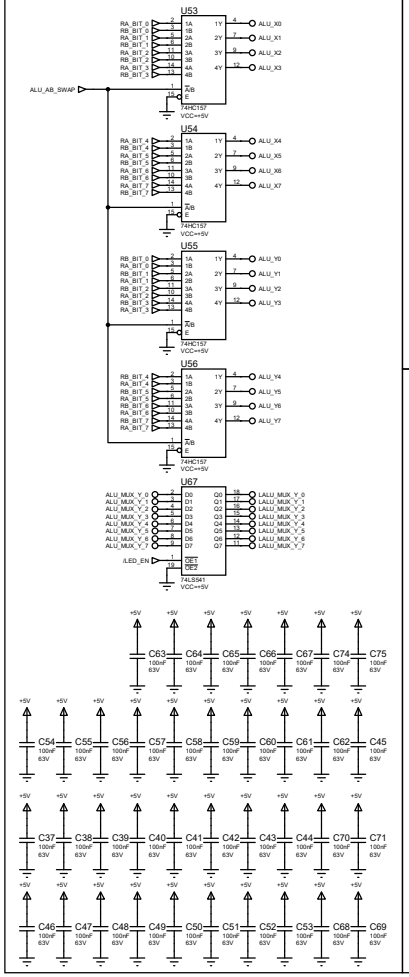




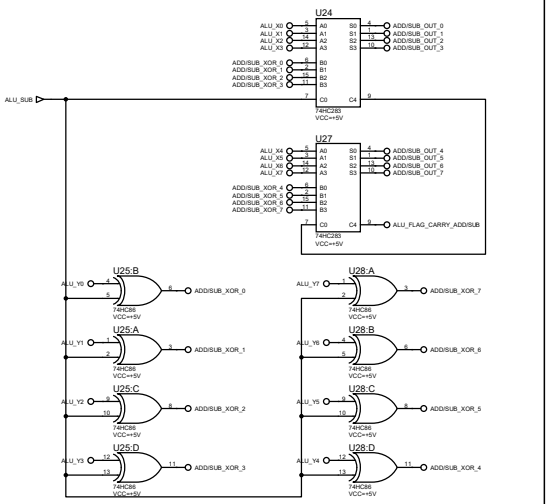
PROJECT: **Wall - 8bits Computer**		DATE: **07/08/2026**
PAGE: **IR Instruction Register**		
FILE NAME: **Hardware.pdsprj**		PAGE: **5 of 17**
DRAWN BY: **Vinicius Jesus**	REV: **1.0**	

ALU
Arithmetic Logic Unit

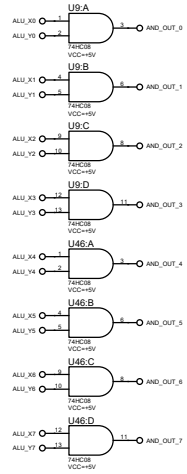
MUX



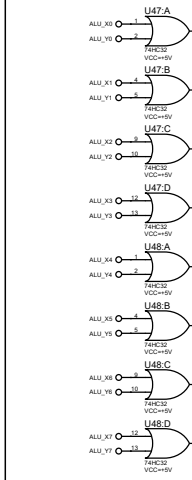
ADD/SUB X +/- Y 8Bits



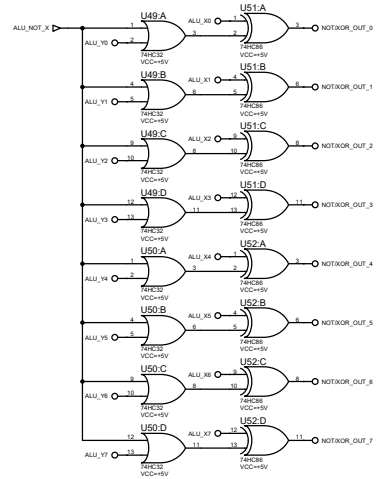
AND X - Y 8Bits



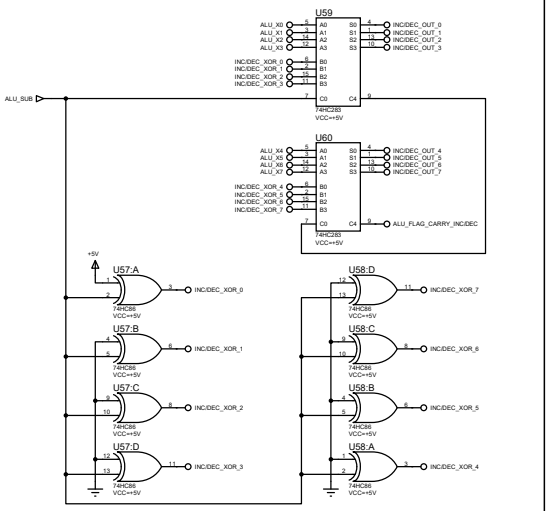
OR X - Y 8Bits



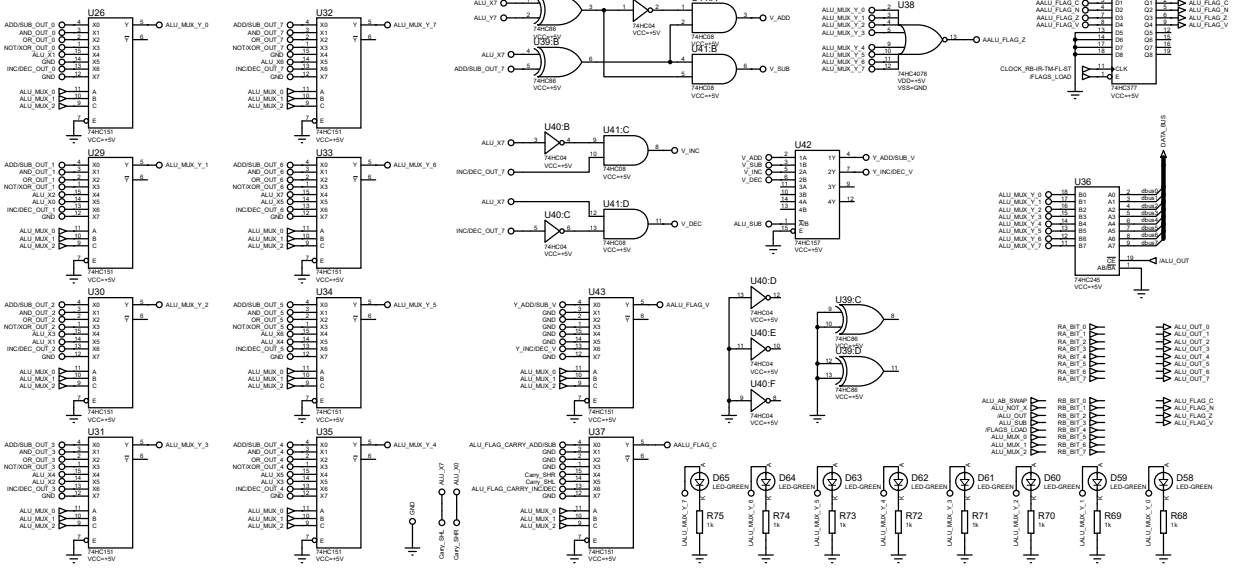
NOT/XOR X XOR Y NOT X 8Bits



INC/DEC X 8Bits



ADD/SUB A + B 8Bits



---------- ALU WALL-8 ----------

ENTRADAS DE DADOS:
RA_BIT_0 ... RA_BIT_7 = Registrador A
RB_BIT_0 ... RB_BIT_7 = Registrador B

BIT 0 = menos significativo
BIT 7 = mais significativo

SELETOR DE ENTRADA A/B:

ALU_AB_SWAP = 0:
ALU_X = RA
ALU_Y = RB

ALU_AB_SWAP = 1:
ALU_X = RB
ALU_Y = RA

ENTRADAS DE CONTROLE:

ALU_MUX_0
ALU_MUX_1
ALU_MUX_2

ALU_NOT_X:
0 = XOR -> ALU_X XOR ALU_Y
1 = NOT -> NOT ALU_X

/ALU_ADD_SUB:
0 = ADD / INC
1 = SUB / DEC

/ALU_BUS:
0 = ALU_OUT habilitada no barramento
1 = ALU_OUT isolada do barramento

TABELA DO ALU_MUX:

ALU_MUX_2 ALU_MUX_1 ALU_MUX_0

000 = ADD/SUB
/ALU_ADD_SUB = 0 -> ALU_X + ALU_Y
/ALU_ADD_SUB = 1 -> ALU_X - ALU_Y

001 = AND
ALU_X AND ALU_Y

010 = OR
ALU_X OR ALU_Y

011 = XOR/NOT
ALU_NOT_X = 0 -> ALU_X XOR ALU_Y
ALU_NOT_X = 1 -> NOT ALU_X

100 = SHR
ALU_OUT = ALU_X >> 1
Carry = ALU_X0

101 = SHL
ALU_OUT = ALU_X << 1
Carry = ALU_X7

110 = INC/DEC
/ALU_ADD_SUB = 0 -> ALU_X + 1
/ALU_ADD_SUB = 1 -> ALU_X - 1

111 = ZERO
ALU_OUT = 00000000

FLAGS:

ALU_FLAG_C:
Carry / Not Borrow / bit deslocado

ALU_FLAG_Z:
1 se ALU_OUT = 00000000
0 se ALU_OUT tiver qualquer bit em 1

ALU_FLAG_N:
ALU_FLAG_N = ALU_OUT_7

ALU_FLAG_V:
Overflow em operação com sinal
Usada em ADD/SUB e INC/DEC
Nas demais operações = 0

CARRY POR OPERAÇÃO:

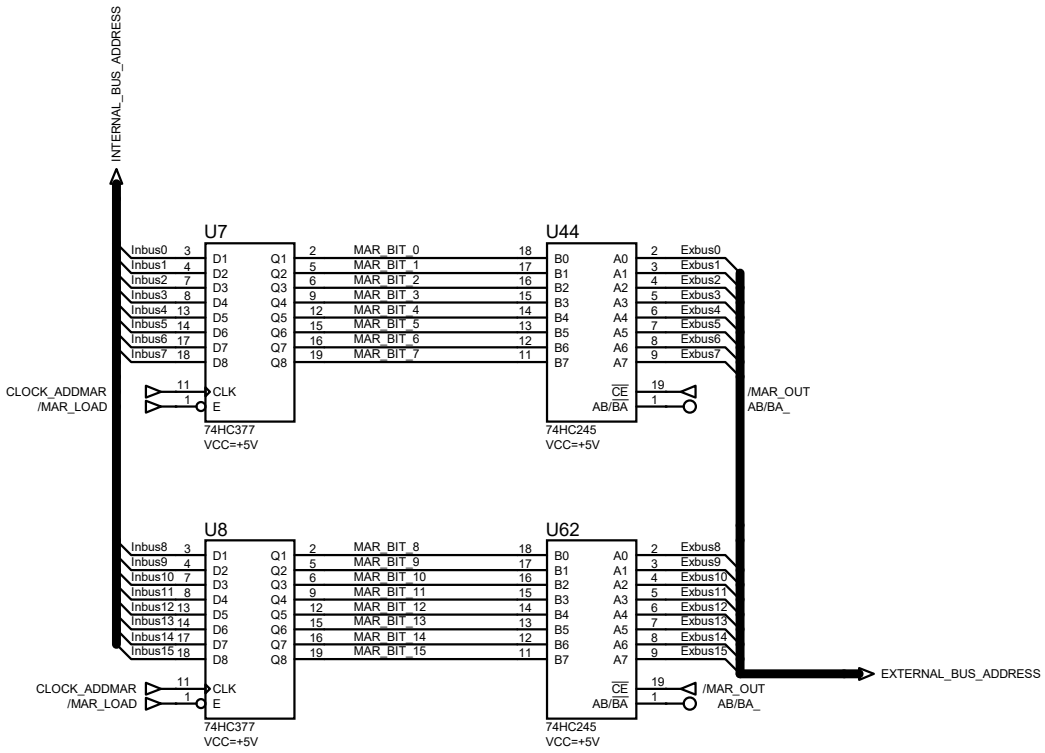
000 ADD/SUB = Carry do ADD/SUB
001 AND = 0
010 OR = 0
011 XOR/NOT = 0
100 SHR = ALU_X0
101 SHL = ALU_X7
110 INC/DEC = Carry do INC/DEC
111 ZERO = 0

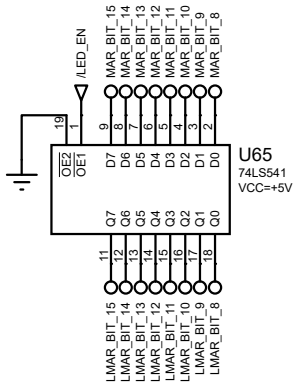
OVERFLOW V POR OPERAÇÃO:

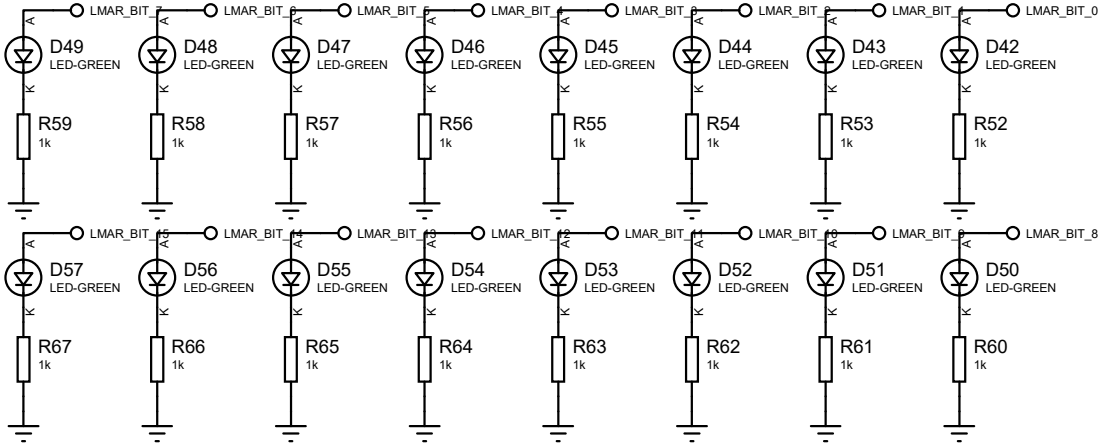
000 ADD/SUB = V do ADD/SUB
001 AND = 0
010 OR = 0
011 XOR/NOT = 0
100 SHR = 0
101 SHL = 0
110 INC/DEC = V do INC/DEC
111 ZERO = 0

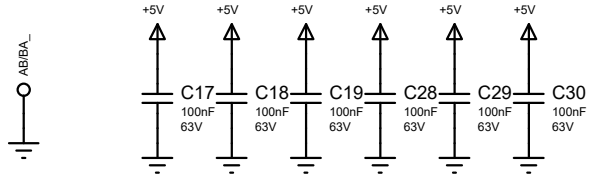
PROJECT: **Wall - 8bits Computer**	DATE: **07/08/2026**
PAGE: **ALU**	PAGE: **6 of 17**
FILE NAME: **Hardware.pdsprj**	
DRAWN BY: **Vinicius Jesus**	REV: **1.0**

MAR - Memory Address Register







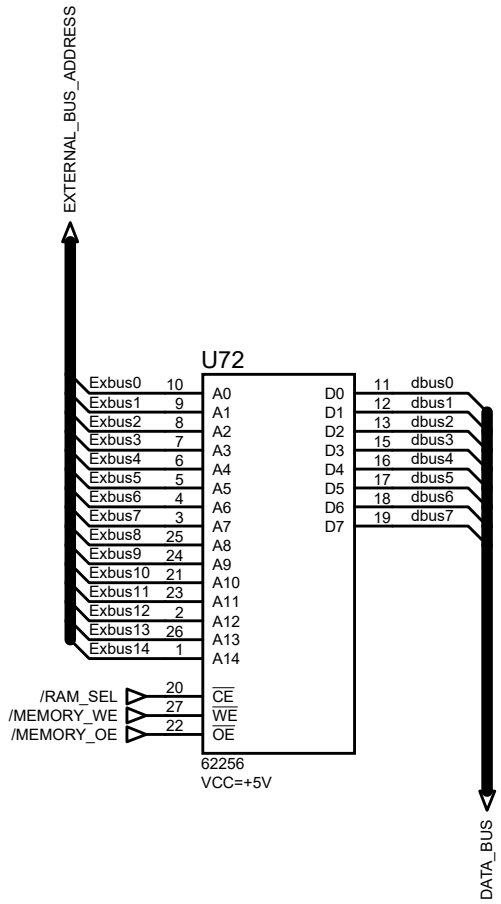


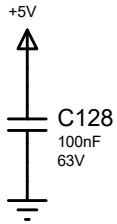
PROJECT: **Wall - 8bits Computer**		DATE: **07/08/2026**
PAGE: **MAR**		
FILE NAME: **Hardware.pdsprj**		PAGE: **7 of 17**
DRAWN BY: **Vinicius Jesus**	REV: **1.0**	

WALL-8 RAM MEMORY

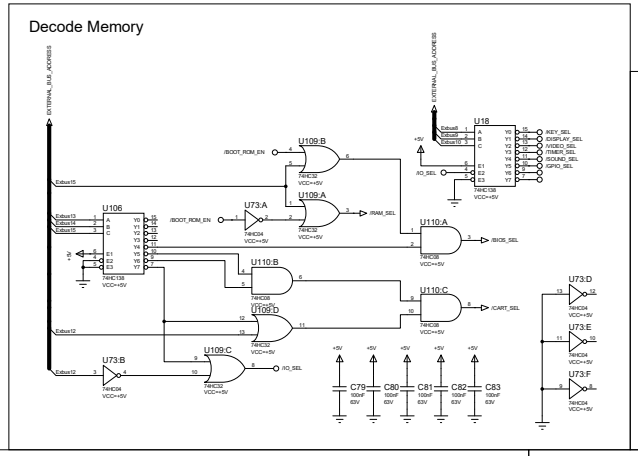
ADDRESS BUS: 16 bits
DATA BUS: 8 bits

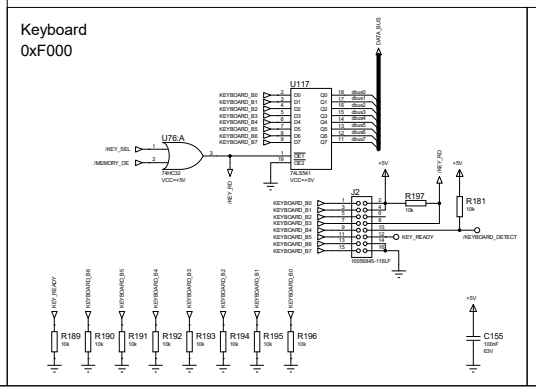
0x0000 - 0x7FFF RAM 32 KB 32768 bytes

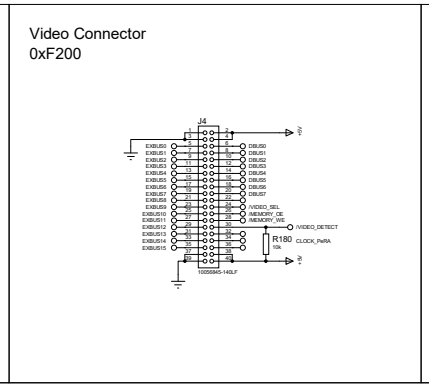


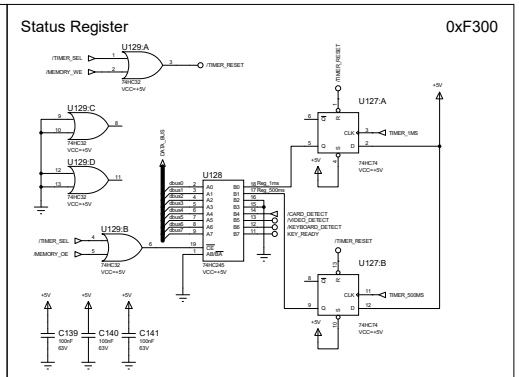


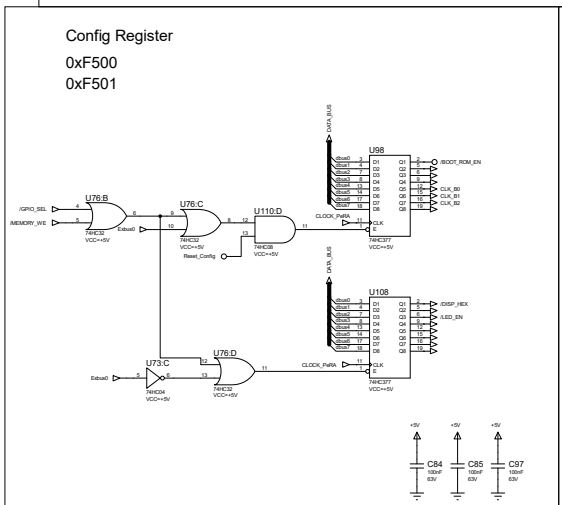
PROJECT: **Wall - 8bits Computer**		DATE: **07/08/2026**
PAGE: **RAM Memory**		
FILE NAME: **Hardware.pdsprj**		PAGE: **8 of 17**
DRAWN BY: **Vinicius Jesus**	REV: **1.0**	

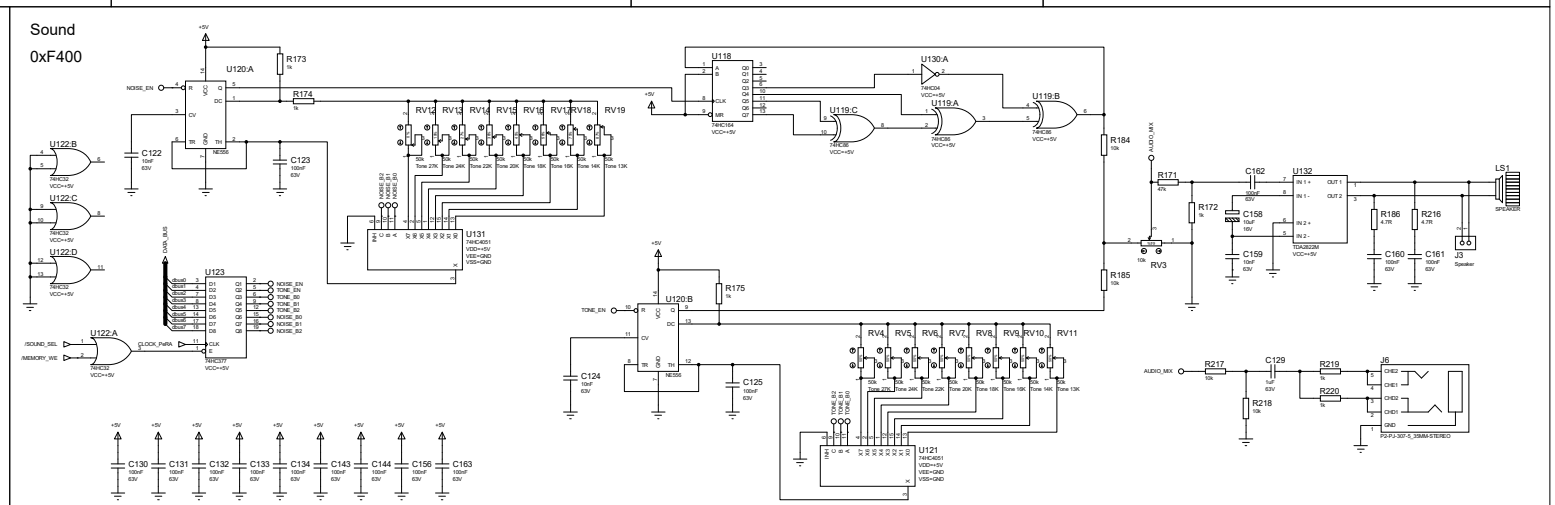












WALL-8 I/O MAP - 74HC138 SUB-DECODE

I/O = 0xF000-0xFFFF
BIN = 1111 xCBA xxxx xxxx

74HC138:
A = A8
B = A9
C = A10

CBA / A10 A9 A8:

000 -> Y0 /KEY_SEL
HEX: 0xF000-0xF0FF and 0xF800-0xF8FF
BIN: 1111 x000 xxxx xxxx

001 -> Y1 /DISPLAY_SEL
HEX: 0xF100-0xF1FF and 0xF900-0xF9FF
BIN: 1111 x001 xxxx xxxx

0xF100 = DISPLAY_LOW
0xF101 = DISPLAY_HIGH

010 -> Y2 /VIDEO_SEL
HEX: 0xF200-0xF2FF and 0xFA00-0xFAFF
BIN: 1111 x010 xxxx xxxx

011 -> Y3 /TIMER_SEL
HEX: 0xF300-0xF3FF and 0xFB00-0xFBFF
BIN: 1111 x011 xxxx xxxx

100 -> Y4 /SOUND_SEL
HEX: 0xF400-0xF4FF and 0xFC00-0xFCFF
BIN: 1111 x100 xxxx xxxx

SOUND_SEL

0xF400 = Bits de config Sound

BIT 0 = /NOISE_EN
BIT 1 = /TONE_EN
BIT 2 = NC
BIT 3 = NC
BIT 4 = TONE_B0
BIT 5 = TONE_B1
BIT 6 = TONE_B2
BIT 7 = NC

101 -> Y5 /GPIO_SEL
HEX: 0xF500-0xF5FF and 0xFD00-0xFDFF
BIN: 1111 x101 xxxx xxxx

0xF500 = Bits de config CPU

BIT 0 = /BOOT_ROM_EN
BIT 1 = NC
BIT 2 = NC
BIT 3 = NC
BIT 4 = /CLK_B0
BIT 5 = /CLK_B1
BIT 6 = /CLK_B2
BIT 7 = NC

0xF501 = Bits de config Perifericos

BIT 0 = /DISP_HEX
BIT 1 = NC
BIT 2 = /LED_EN
BIT 3 = NC
BIT 4 = NC
BIT 5 = NC
BIT 6 = NC
BIT 7 = NC

110 -> Y6 Não Usado
HEX: 0xF600-0xF6FF and 0xFE00-0xFEFF
BIN: 1111 x110 xxxx xxxx

111 -> Y7 Não Usado
HEX: 0xF700-0xF7FF and 0xFF00-0xFFFF
BIN: 1111 x111 xxxx xxxx

WALL-8 MEMORY MAP - 64 KB ADDRESS SPACE

ADDRESS BUS: 16 bits
DATA BUS: 8 bits

0x0000 - 0x7FFF RAM 32 KB 32768 bytes
0x8000 - 0x9FFF BIOS/ROM 8 KB 8192 bytes
0xA000 - 0xEFFF CART 20 KB 20480 bytes
0xF000 - 0xFFFF I/O 4 KB 4096 bytes

TOTAL: 64 KB = 65536 bytes

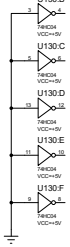
BOOT MODE:

When BOOT_ROM_EN = 0:
0x0000 - 0x7FFF BIOS mirror / boot overlay
0x8000 - 0x9FFF BIOS fixed

When BOOT_ROM_EN = 1:
0x0000 - 0x7FFF RAM
0x8000 - 0x9FFF BIOS fixed

DECODE:

RAM_SEL = 0x0000 - 0x7FFF
BIOS_SEL = 0x8000 - 0x9FFF
CART_SEL = 0xA000 - 0xEFFF
IO_SEL = 0xF000 - 0xFFFF

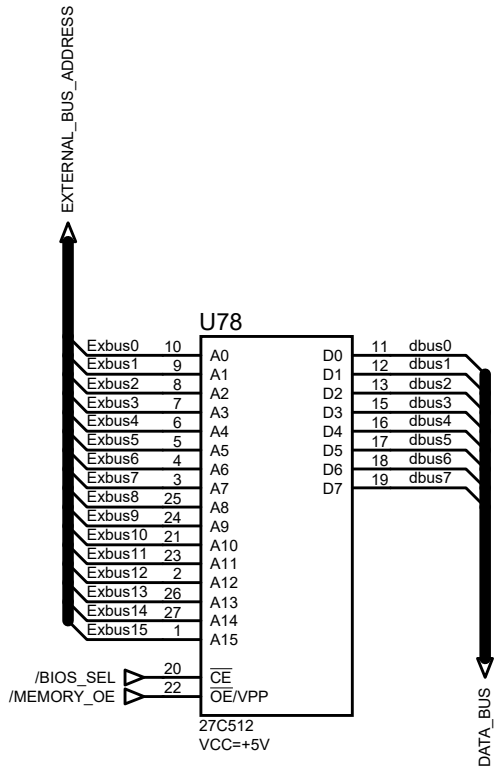


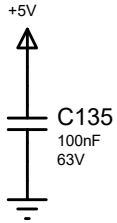
PROJECT: Wall - 8bits Computer	DATE: 07/08/2026
PAGE: Decode Memory	
FILE NAME: Hardware.pdsprj	PAGE: 5 of 17
DRAWN BY: Vinicius Jesus	REV: 1.0

WALL-8 ROM MEMORY

ADDRESS BUS: 16 bits
DATA BUS: 8 bits

0x8000 - 0x9FFF BIOS/ROM 8 KB 8192 bytes



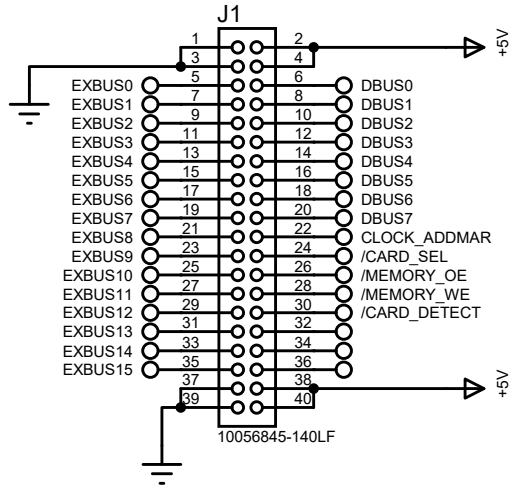


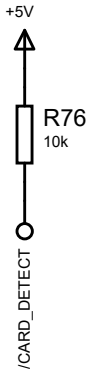
PROJECT: **Wall - 8bits Computer**	DATE: **07/08/2026**
PAGE: **ROM/BIOS Memory**	
FILE NAME: **Hardware.pdsprj**	PAGE: **10 of 17**
DRAWN BY: **Vinicius Jesus** REV: **1.0**	

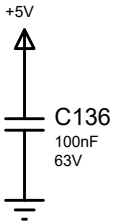
WALL-8 CARD MEMORY

ADDRESS BUS: 16 bits
DATA BUS: 8 bits

0xA000 - 0xEFFF CARD 20 KB 20.480 bytes

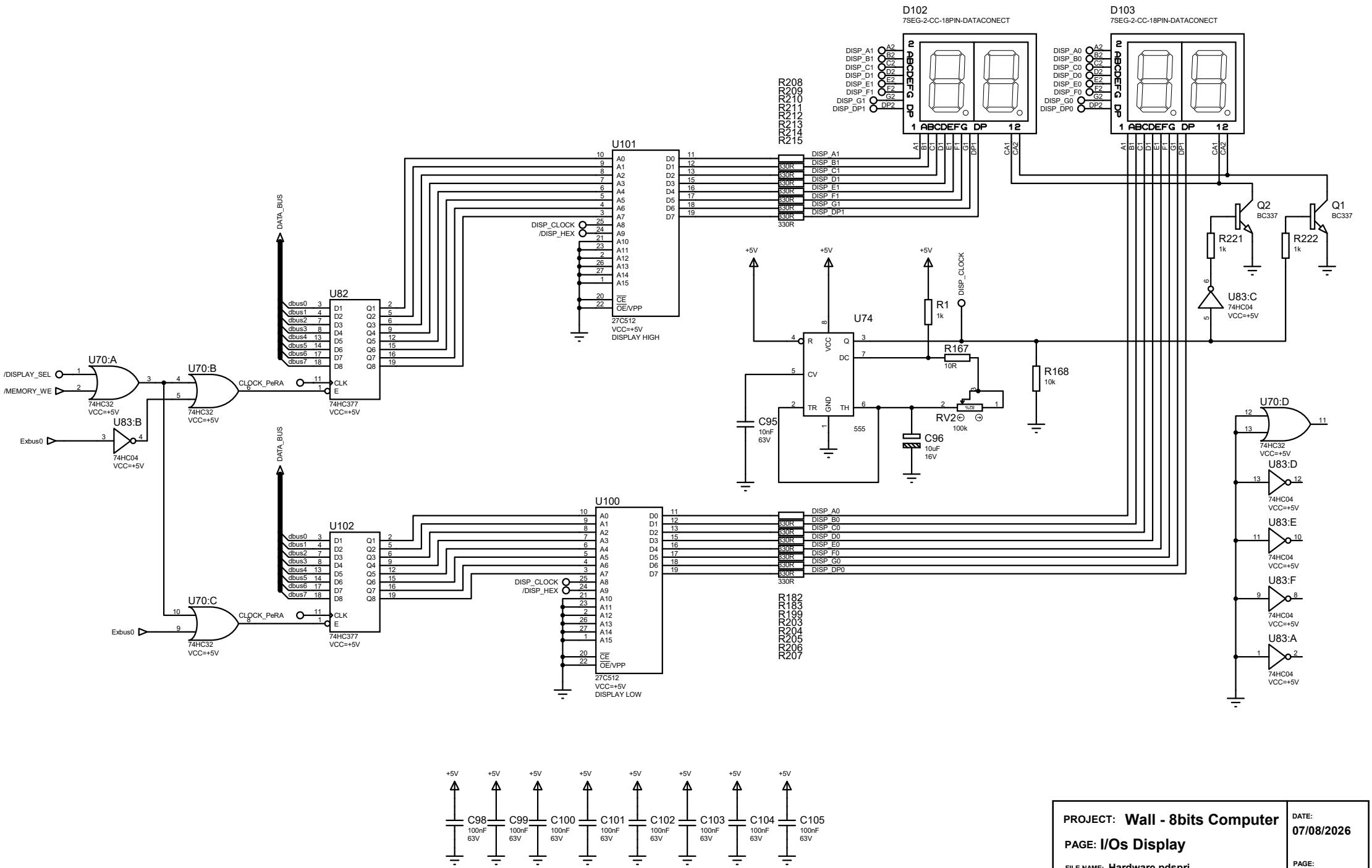






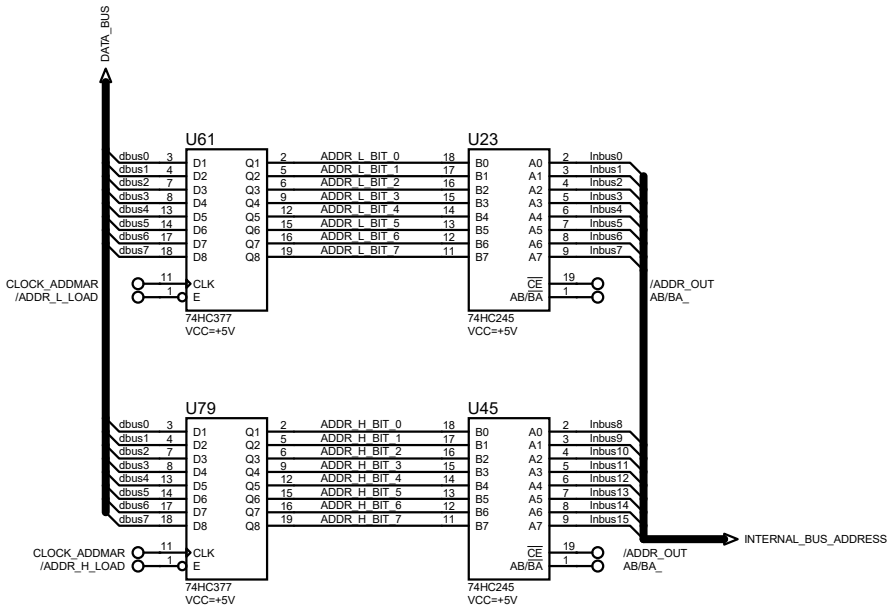
PROJECT: Wall - 8bits Computer	**DATE:** 07/08/2026
PAGE: CARD Memory	
FILE NAME: Hardware.pdsprj	**PAGE:** 11 of 17
DRAWN BY: Vinicius Jesus **REV:** 1.0	

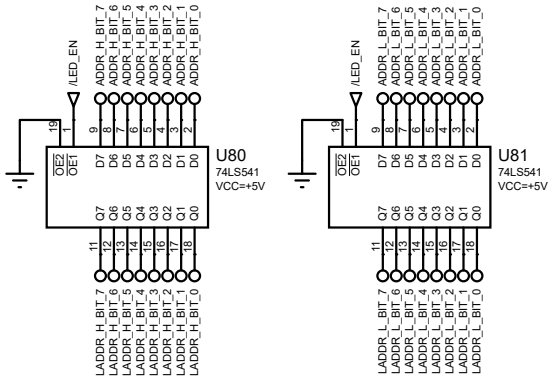
DISPLAY HEX

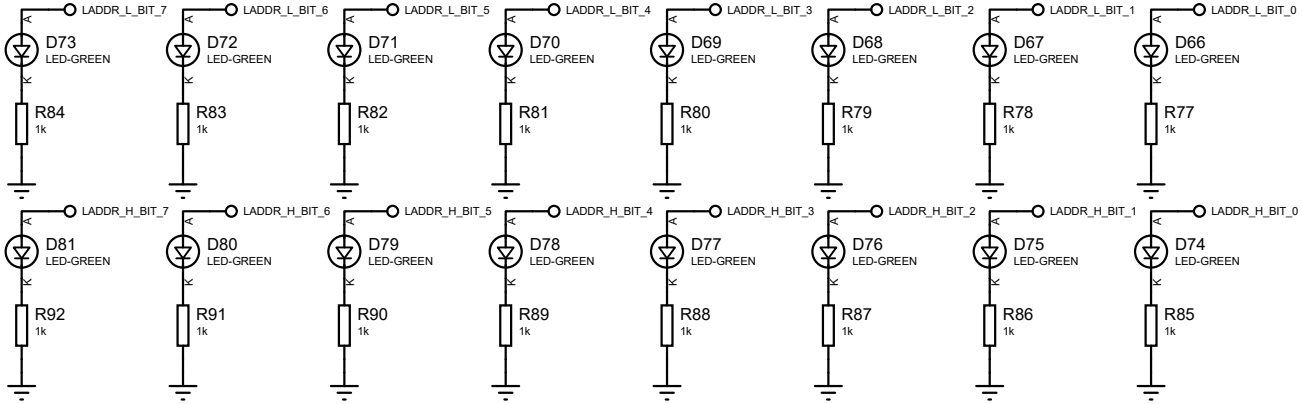


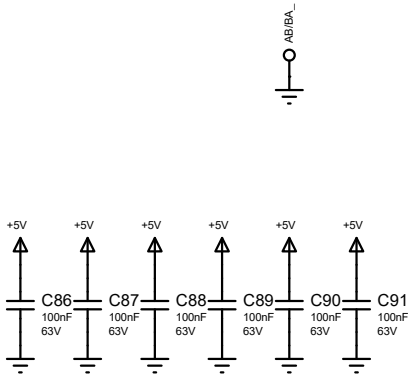
PROJECT: **Wall - 8bits Computer**	DATE: **07/08/2026**
PAGE: **I/Os Display**	PAGE: **12 of 17**
FILE NAME: **Hardware.pdsprj**	
DRAWN BY: **Vinicius Jesus**	REV: **1.0**

ADDR - Address Register







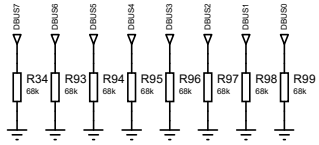


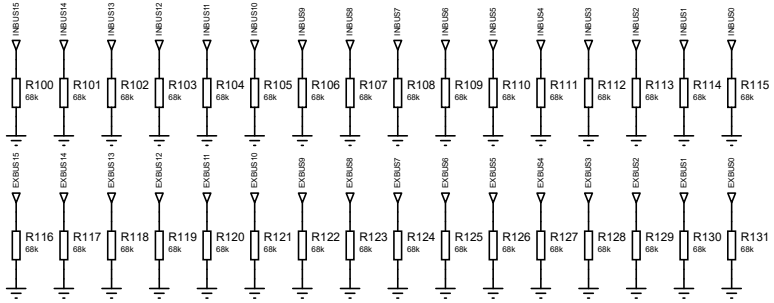
PROJECT: **Wall - 8bits Computer**	DATE: **07/08/2026**
PAGE: **ADDR**	PAGE: **13 of 17**
FILE NAME: **Hardware.pdsprj**	
DRAWN BY: **Vinicius Jesus** REV: **1.0**	

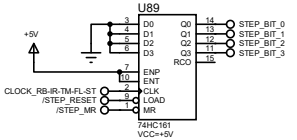
WALL-8 UNIT CONTROL

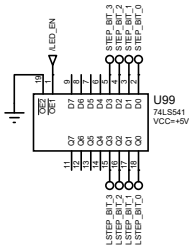
Valores de NEUTRO , (TUDO DESABILITADO)

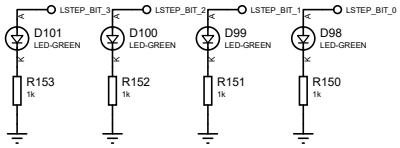
ROM 1 0xFE
ROM 2 0X1F
ROM 3 0xF8
ROM 4 0xDF
ROM 5 0x6F

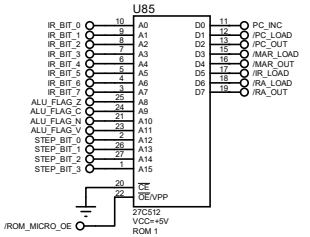


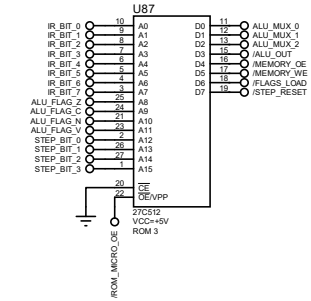


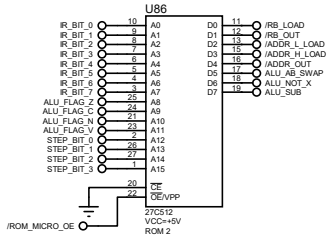


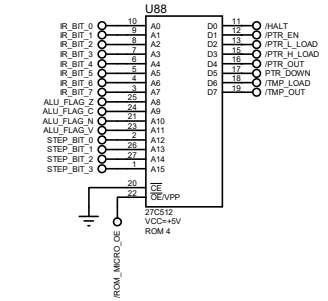


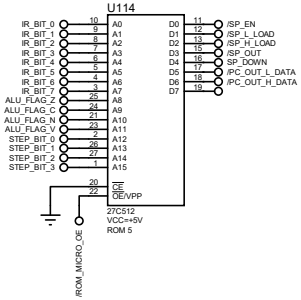


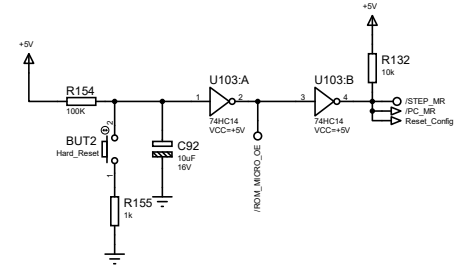


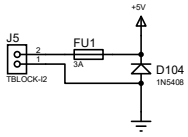


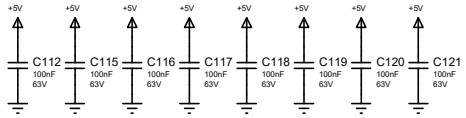


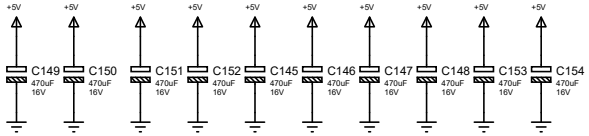


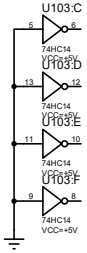


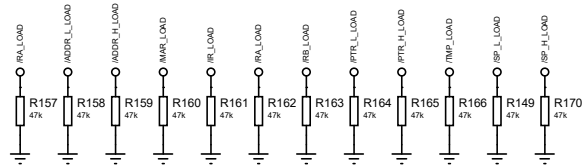








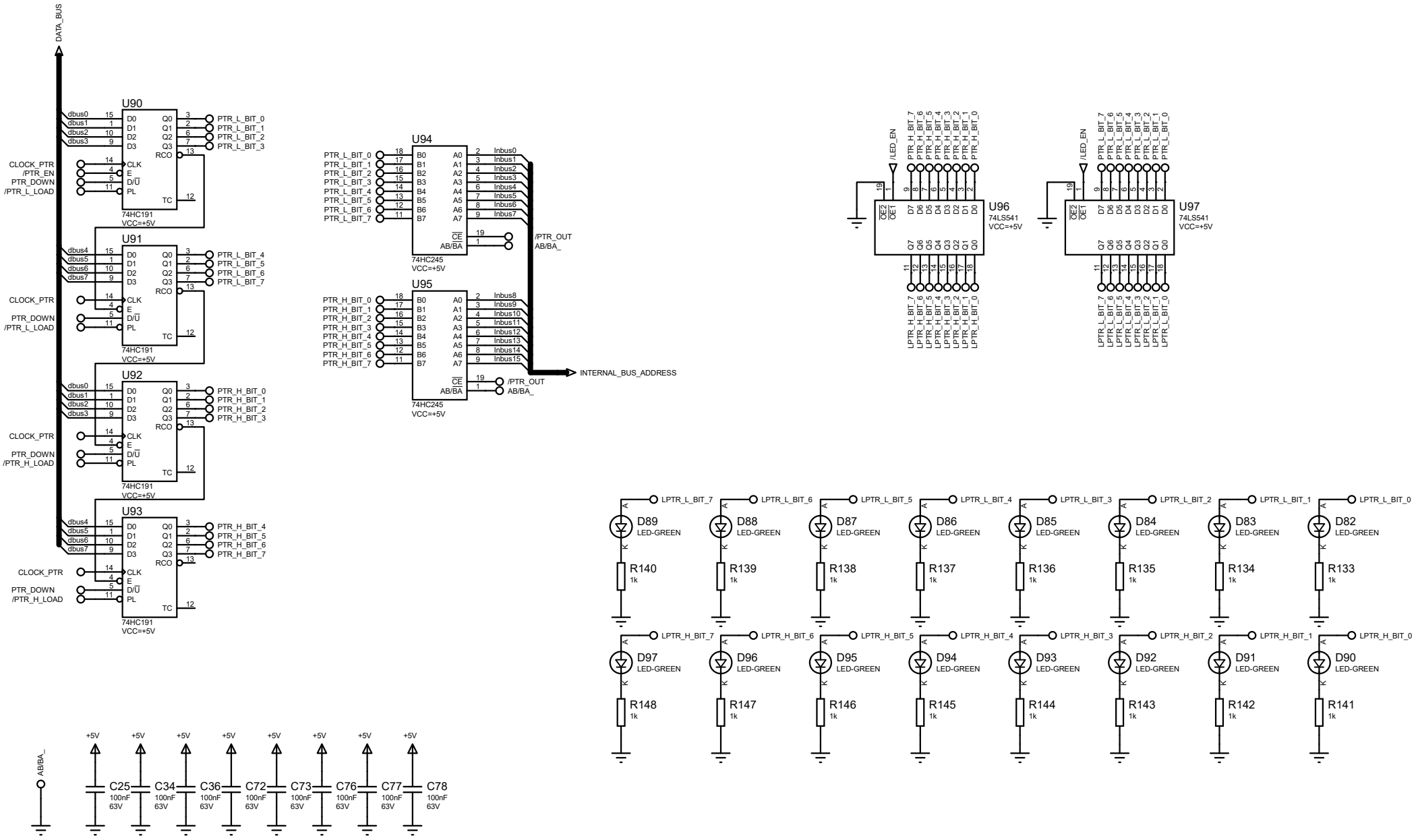






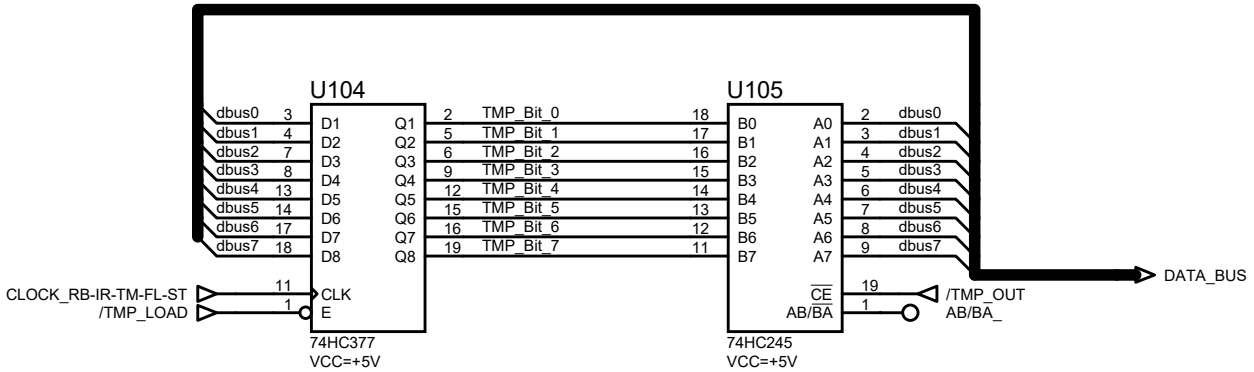
PROJECT: **Wall - 8bits Computer**	DATE: **07/08/2026**
PAGE: **Unit Control**	PAGE: **14 of 17**
FILE NAME: **Hardware.pdsprj**	
DRAWN BY: **Vinicius Jesus** REV: **1.0**	

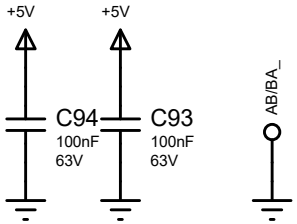
PTR - Pointer Register



PROJECT: **Wall - 8bits Computer**	DATE: **07/08/2026**
PAGE: **PTR**	
FILE NAME: **Hardware.pdsprj**	PAGE: **15 of 17**
DRAWN BY: **Vinicius Jesus** REV: **1.0**	

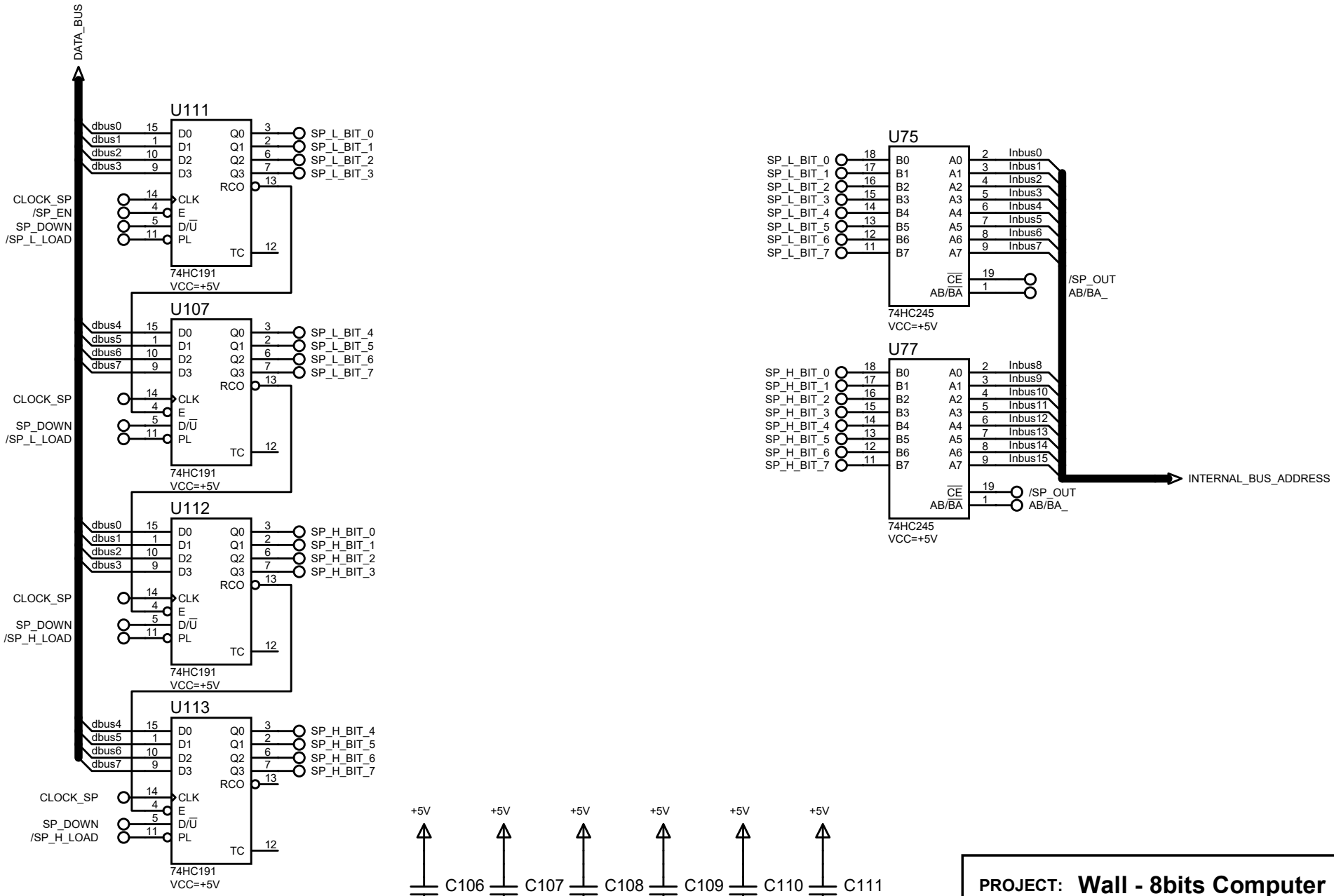
TMP - Temporary Register





PROJECT: **Wall - 8bits Computer**		DATE: **07/08/2026**
PAGE: **Registrador TMP**		
FILE NAME: **Hardware.pdsprj**		PAGE: **16 of 17**
DRAWN BY: **Vinicius Jesus**	REV: **1.0**	

SP - Stack Point



PROJECT: Wall - 8bits Computer	**DATE:** 07/08/2026
PAGE: SP - Stack Point	
FILE NAME: Hardware.pdsprj	**PAGE:** 17 of 17
DRAWN BY: Vinicius Jesus **REV:** 1.0	